

# Sequential Fine-Tuning Order Fingerprints

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Can the final model reveal which task it learned last?

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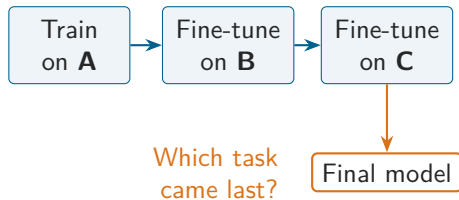
# Research question

A network is trained on tasks one after another.  
Each new task changes the weights.

**Can we predict the final learned task?**

**Can we predict the full training order?**

**Explore catastrophic forgetting**



## Setup: three small classification tasks

Each task is a 5-class problem built from CIFAR-100 classes. All tasks share one ResNet18 backbone; every task has its own 5-class head.

| Task | Theme                       | Classes                                       |
|------|-----------------------------|-----------------------------------------------|
| A    | Aquatic animals             | beaver, dolphin, otter, seal, whale           |
| B    | Electronic / office devices | clock, keyboard, lamp, telephone, television  |
| C    | Vehicles                    | bicycle, bus, motorcycle, pickup truck, train |

Six task orders are tested: all permutations of A, B, C.

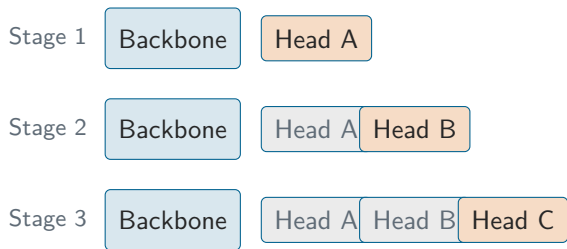
The **actual last task** of an order is simply its final element.

# How the sequential models are trained

For an order such as  $A \rightarrow B \rightarrow C$ :

1. Train backbone and A-head on A.
2. Freeze the A-head, add a B-head, train on B.
3. Freeze A- and B-heads, add a C-head, train on C.

A checkpoint is saved after every stage; the last one is the **final sequential model**.



Orange = being trained. Grey = frozen head.

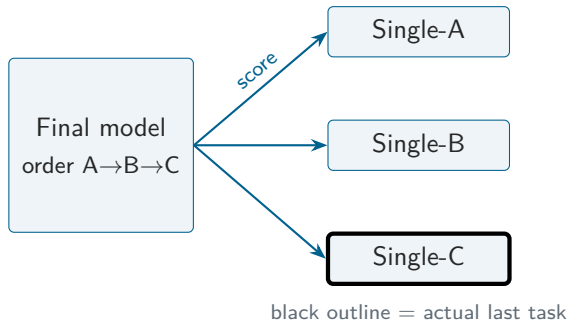
# The fingerprint test

**Reference models:** three models, each trained only on A, B or C (Single-A, Single-B, Single-C).

**Test:** compare the final sequential model with each reference, and pick the reference that looks most similar. That task is the **predicted last task**.

A prediction is correct if it equals the last element of the order.

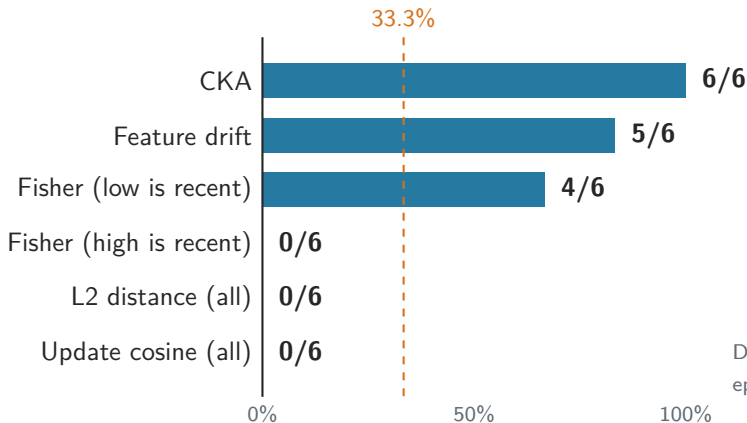
Random guessing gives  $1/3 = 33.3\%$ .



# Methods overview: predictors and diagnostics

| Method                        | What is compared                           | Prediction rule             |
|-------------------------------|--------------------------------------------|-----------------------------|
| <b>Fingerprint predictors</b> |                                            |                             |
| Weight distance (L2)          | Final model vs. each reference             | Smallest distance           |
| CKA                           | Feature structure on a fixed image set     | Highest similarity          |
| Feature drift                 | Feature vectors, image by image            | Smallest drift              |
| Weight matching               | Weights after undoing neuron reordering    | Smallest L2, highest cosine |
| Fisher trace                  | Curvature of the final model alone         | Both directions tested      |
| <b>Forgetting diagnostics</b> |                                            |                             |
| Forgetting                    | Accuracy after learning vs. final accuracy | -                           |
| Loss barrier                  | Loss along path reference → final model    | -                           |
| Jacobian sensitivity          | Logit gradient w.r.t. the input image      | -                           |

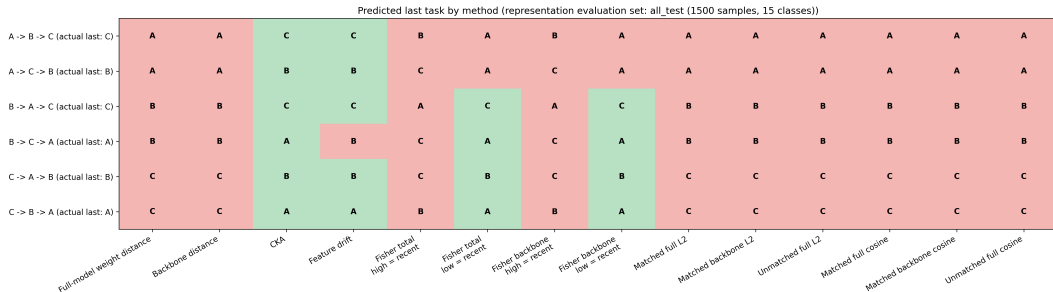
## Result: how often is the last task predicted correctly?



- CKA is right in every order.
- Feature drift misses one.
- Fisher works only as “lowest = most recent” (total and backbone).
- L2 and cosine are wrong in every order, also after matching.

Dashed: 33.3% baseline. 20 epochs/task, all\_test, seed 42.

# Prediction matrix: every order, every method

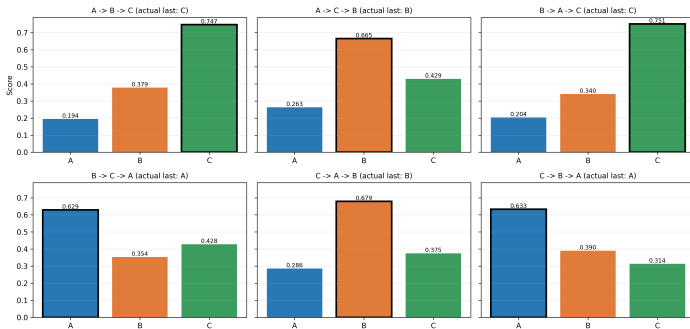


Rows are orders, columns are methods, cells show the predicted task. **Green** is correct, **red** is wrong.

In every L2 and cosine column the predicted task is the **first** task of the order, never the last.

# CKA separates the last task clearly

Asks: is the structure of the representation similar?  
Highest CKA predicts the last task.



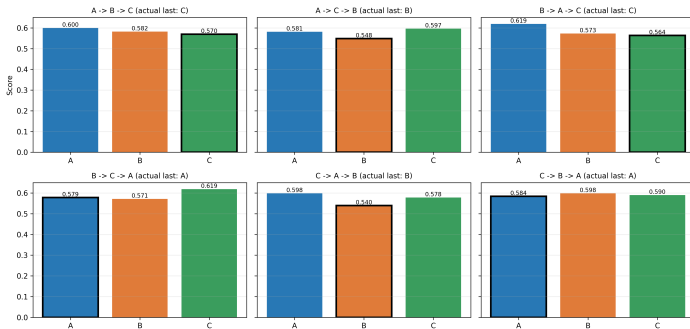
Bars show CKA between the final model and Single-A, B, C. Outline marks the actual last task.

- The outlined bar is the highest in all six orders.
- Last-task CKA lies between 0.63 and 0.75.
- The other two references lie between 0.19 and 0.43.

The gap is large, not a near tie.

# Feature drift

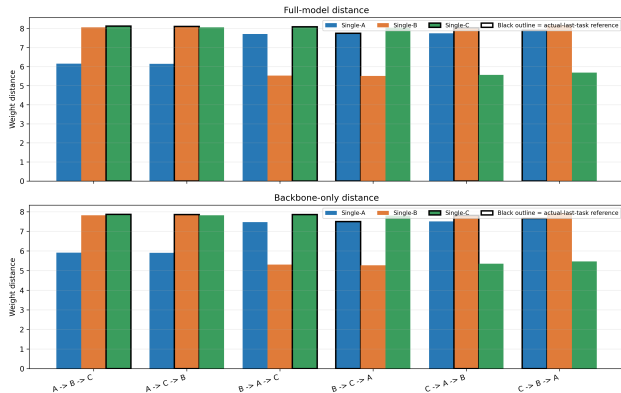
Asks: did the individual feature vectors move? Lowest drift predicts the last task.



Lower drift means more similar features. Outline marks the actual last task.

- The outlined bar is the lowest in 5 of 6 orders.
- The miss is  $B \rightarrow C \rightarrow A$ : Single-B (0.571) is just below Single-A (0.579).
- All values lie between 0.54 and 0.62, so margins are small compared with CKA.

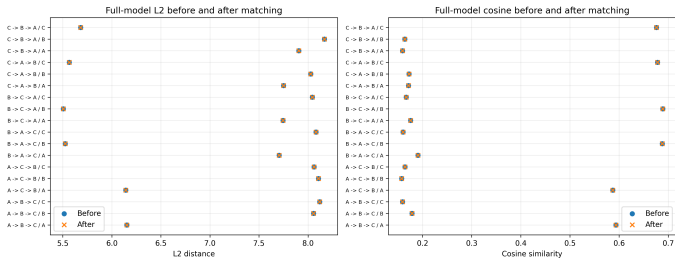
# Weight distance points at the first task



L2 distance from each final model to the three references, for the full model (top) and the backbone (bottom). Outline marks the actual last task.

- In every order the **closest** reference is the **first** task.
- The last task is never the closest: 0 of 6.
- Full model and backbone give the same picture.

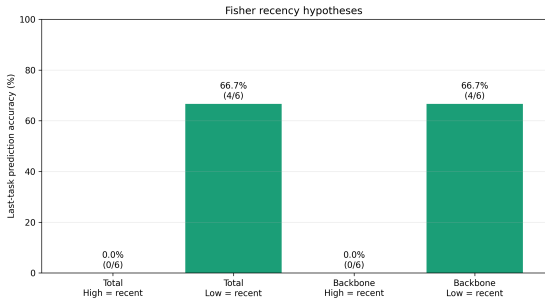
# Weight matching changes nothing



Each row is one final model and one reference. Dots: before matching. Crosses: after.

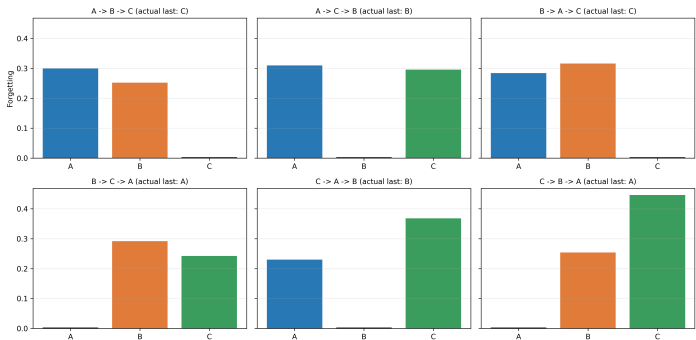
- Dots and crosses coincide everywhere: distances did not change.
- This fits the expected identity permutation with a shared initialization.
- The cosine of weight updates is highest for the first task (about 0.59 to 0.69, others 0.16 to 0.19), so it fails like L2.

# Fisher: lowest trace marks the last task in 4 of 6



- `low_is_recent`: 4/6 for both total and backbone Fisher.
- `high_is_recent`: 0/6.
- The two misses are the orders that start with A. There, the first task A has the lowest trace and is predicted.
- So this method predicts A in 4 of 6 orders, and A is last in only 2.
- Full training order: 1/6 correct, chance is 1/6.

# Forgetting



$\text{forgetting}(T) =$   
acc. after learning  $T$  –  
final acc. on  $T$

- Earlier tasks lose about 0.23 to 0.45 accuracy.
- The last task shows zero forgetting in every order.
- That is by construction: for the last task, “after learning” and “final” are the same checkpoint.

So forgetting gives no independent evidence of a fingerprint.

# Hypotheses and outcome

|    | Hypothesis                              | Outcome so far                                                             |
|----|-----------------------------------------|----------------------------------------------------------------------------|
| H1 | Can we predict the final learned task?  | CKA 6/6, feature drift 5/6, Fisher (low is recent) 4/6. L2 and cosine 0/6. |
| H2 | Can we predict the full training order? | In combination of CKA and weight distance                                  |
| H3 | Explore catastrophic forgetting         | ...                                                                        |

# Limitations

- **One training run.** No estimate of run-to-run variation.
- **Only 3 Tasks per Order.**
- **Fisher predictions are concentrated on A** (4 of 6 orders).
- **Small tasks.** 5 classes each, from CIFAR-100; the tasks may be too similar or too different.

# Conclusions and next steps

## Conclusions

- The last task is detectable in the **representations** (CKA, feature drift) and, less reliably, in the **curvature** (Fisher).
- Weight distance and update cosine point at the first task.
- Forgetting adds no independent evidence.